HALEEMA TALLAT

AI Engineer · Evaluations, agents, LLM inference
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AI SAFETY & EVALUATIONS

ConspiracyBench — Stationhouse, research collaborator · judge study 2026 – Ongoing

- Led the project's judge selection, a scalable-oversight question: built a 150-reply blind validity set with pre-registered scoring, coordinated 4 labellers, and compared 9 judges plus jury and cascade designs. The incumbent judge called divergence at nearly twice the human rate (43% vs 23%); the team's vote was based on these results.
- Found the headline divergence rate was inflated from about 20% to 49%: Qwen 3.5 4B spent its token budget on reasoning and returned empty answers on half its turns, which the judge still graded. Fixed the scorer to mark them invalid; another contributor reproduced the issue on two more judges.
- Fixed a parser bug that read negated labels ("does not diverge") as verdicts, and proposed the study's reliability bar.
- Built a log-analysis tool that shows where a run's time goes and checks byte for byte that speed changes leave prompts unchanged; it traced 51 timed-out requests per run to a concurrency default.

LeakBench — github.com/haleematallat/leakbench 2026

- Built a 30-task Inspect benchmark that tests whether AI agents can find and fix the bugs that make an ML evaluation report the wrong number, with 5 correct pipelines to catch over-editing and hidden checks in Docker sandboxes.
- Made every task verifiable and hard to reward-hack before any model ran: 120 no-model checks show each is solvable and fails if data is edited or tests are faked. Fixed ranges that rejected valid fixes or passed half-fixes.
- Showed it separates models: Qwen 3.5 9B solved 75% of bug tasks against 29% for the 4B (95% CI +29 to +61 points). Without a hint that the eval might be wrong, the 4B fell to 12%.

Technical AI Safety course — BlueDot Impact Sep 20 – Oct 24, 2026

ARENA curriculum (self-study) 2026

EXPERIENCE

Senior AI Engineer, Agents & LLM Infrastructure — IgniteTech (via Crossover) Jul 2025 – Present

- Designed and built IRIS's first production social engine in ~3 months: 1,000+ customer interactions a day across 10–15 channels on multi-tenant Node.js/TypeScript services on AWS. It became the template for 20 more engines; scaled the platform to 27 on Kafka/KEDA and cut memory 6–10× and startup ~20× by replacing Java monoliths.
- Architected a distributed platform running up to 500 concurrent coding agents: LangGraph on Ray, Pydantic-typed state and tools, Kafka events, Redis and PostgreSQL state, on Kubernetes on AWS.
- Made agent runs measurable by tracing every model and tool call in Langfuse, tracking cost per run and tool-call errors in ClickHouse, and scoring drafted replies by acceptance and edit rates.
- Designed the governed MCP layer for Claude Code, Cursor and our agents (OAuth2, capability controls, human-in-the-loop approval for high-impact actions), and a rule-checking agent on top of it.

Computer Vision Engineer — Brick & Mortar AI Mar – Oct 2025

- Fine-tuned a ResNet-12 few-shot classifier on damaged barcode images, then showed it scored 89.7%, not the reported 46.7%, by fixing split leakage and protocol mismatches in the eval harness.
- Cut identity switches 20% with segmentation-masked ReID; GPU RTSP decoding raised throughput 10–15×.
- Took the tracking pipeline to production: PyTorch to ONNX to TensorRT, a custom Triton ReID backend with a re-tuned matching threshold, and fixes for shape-mismatch and runner-registration failures in the deployed stack.

Computer Vision Engineer — NextBridge Nov 2024 – Mar 2025

- Built a smart wheelchair's WebRTC control stack: signalling server (SDP/ICE), joystick sender, ROS 2 receiver.
- Made the link fail-safe with heartbeats, backoff reconnection and a watchdog that stops the chair on link loss; instrumented round-trip latency.

Machine Learning Engineer — Powersoft19 Nov 2023 – Nov 2024

- Cut field engineers' problem-resolution time 90% with a hybrid RAG system over technical docs, tuned on recall@k.
- Cut fault-detection turnaround from ~48 hours to ~30 minutes by leading 3 engineers to replace manual inspection with anomaly detection on 1TB+ of device telemetry.
- Built an ML experimentation platform with CLI tools, monitoring dashboards and GitHub integration that let 200+ engineers deploy and track experiments, cutting setup from days to hours.

Research Engineer — KICS, Center for Language Engineering, UET Nov 2022 – Nov 2023

- Reached 98% accuracy with a distributed OCR pipeline for historical archives, running at 99.9% uptime.
- Reviewed the literature on OCR techniques for a multilingual LLM setup, and reproduced published RAG methods.
- Was part of the team that built Tajziyakar, an archive-scale OCR pipeline that flags TV news tickers.

EDUCATION & SKILLS

Education: BSc Electrical Engineering, UET Lahore (2023)
Awards: CureMD Master Inventor Award (2022) · COHACK by DevNation hackathon, 1st place (2022)
Evaluations: Inspect, benchmark and task design, LLM-as-a-judge validation, human–AI agreement, paired bootstrap
LLMs & agentic coding: Claude Code, Cursor, Codex, MCP, LangGraph, Anthropic API, LibreChat, RAG
Computer vision: YOLOv8/v11, segmentation, re-identification, face recognition, facial landmarks, zone monitoring
Model serving: vLLM, ONNX, TensorRT, Triton Inference Server, NVIDIA Jetson, CUDA
Cloud & data: AWS (EKS, S3, DynamoDB), Kubernetes, Docker, Kafka, PostgreSQL, MongoDB, Redis, ClickHouse
Languages & ML: Python, TypeScript, C, C++, SQL, Node.js, Next.js, React, PyTorch, JAX, fine-tuning